

Technical Architecture

Student portfolio website (web development assignment)

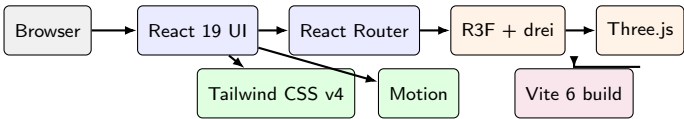
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AIMS assignment context. Student-authored portfolio for a web-development coursework component; this one-pager documents the technical architecture of that deliverable.
<https://rosly-mamekem.vercel.app/>



Purpose. As an AIMS student, I implemented this portfolio to showcase research interests, projects, and credentials. The site is a client-side rendered (CSR) single-page application (SPA) with one extra route for a certifications gallery. Content is authored in TypeScript/React; styling is utility-first; the hero and project sections embed interactive 3D scenes for engagement and demonstration of modern front-end skills.

Presentation layer. React 19 and TypeScript structure the UI into pages (`App.tsx`, `pages/Certifications.tsx`) and components (`Scene`, `ProjectCarousel`, `CipherGate`, etc.). Tailwind CSS v4 (via `@tailwindcss/vite`) supplies design tokens and responsive layout. Motion drives scroll/viewport animations. lucide-react provides icons.

Navigation & state. react-router-dom maps paths (`/`, `/certifications`). Local UI state uses React hooks. Theme (`ThemeContext`) toggles light/dark, persists `portfolio-theme` in `localStorage`, mirrors `html.light` for global CSS overrides, and feeds accent colors into Three.js materials where needed.

3D & media. @react-three/fiber bridges React to Three.js: background particle field (`Scene`), carousel project cards with textures (`ProjectCarousel`), lighting and fog. @react-three/drei adds helpers (`Image`, `Text`, `Grid`, etc.). Raster assets ship as bundled imports (Vite).

Build & delivery. Vite 6 with @vitejs/plugin-react bundles ESM for production (`npm run build → dist/`). Typical deployment is static hosting (e.g. Vercel).

Security / UX extras. A gated section (`CipherGate`) conditionally reveals content after a client-side puzzle. No server is required for the static portfolio shell; optional Node packages in the repo may support separate tooling and are not part of the static export unless explicitly wired.

Concern	Primary technologies
UI framework	React 19, TypeScript
Styling	Tailwind CSS 4 (@tailwindcss/vite)
Routing	react-router-dom 7
Animation	Motion 12
3D	three, @react-three/fiber 9, @react-three/drei 10
Tooling	Vite 6, ESLint-style check via <code>tsc -noEmit</code>

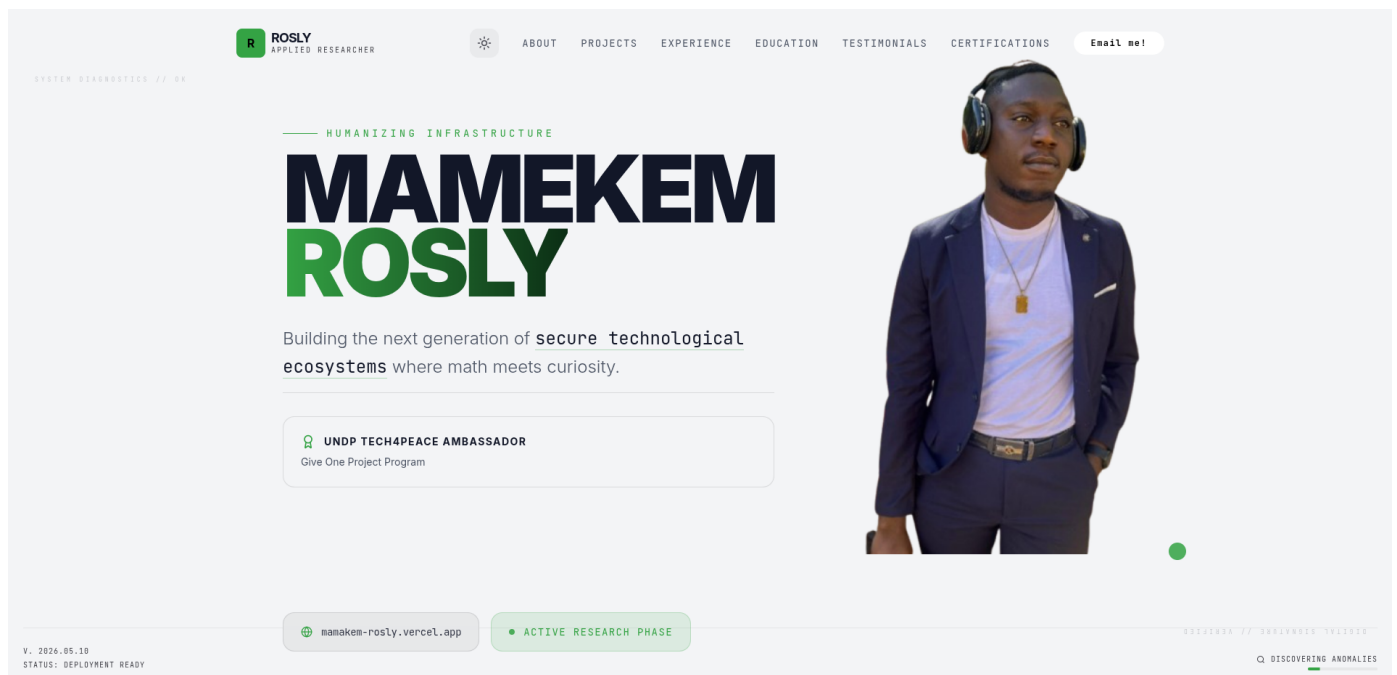


Figure 1:

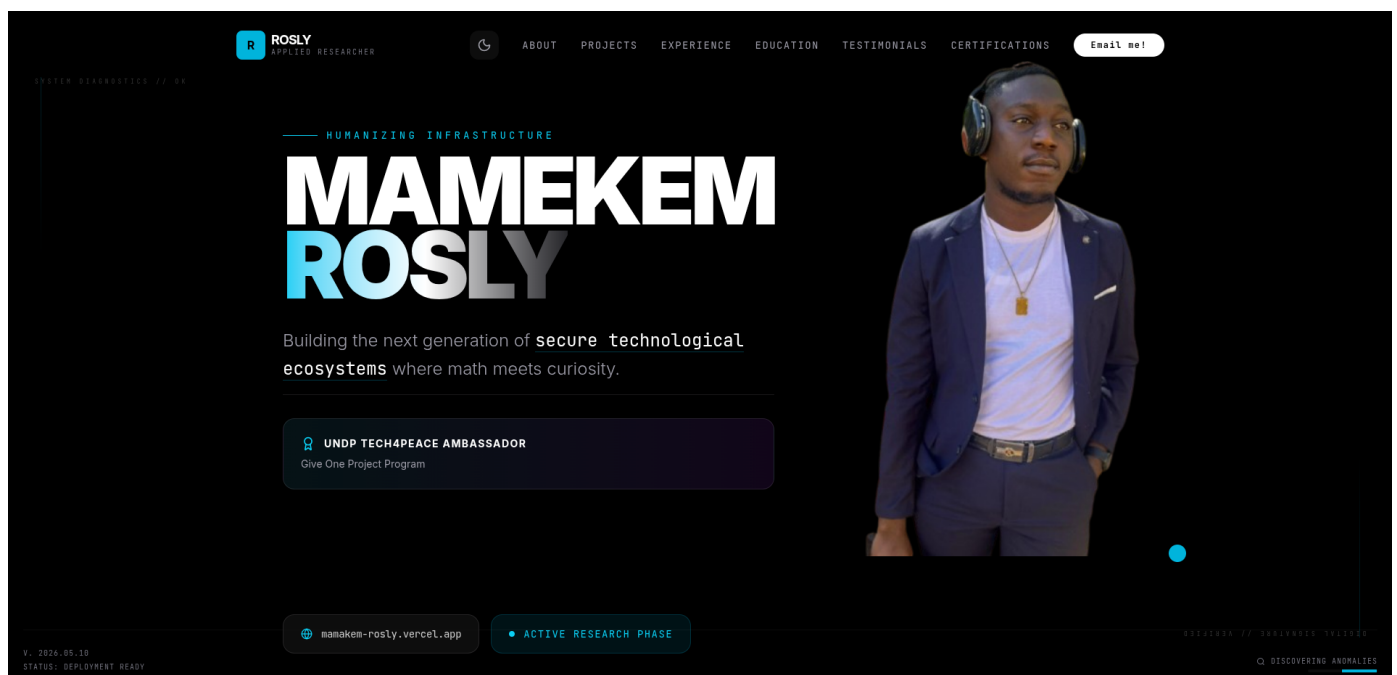


Figure 2: